PICARD GROUP OF A GLUEING OF TWO POINTS: EXACT SEQUENCE WITH KERNEL CONTROLLED BY THE TWO MARKED POINTS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file curves.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the surjectivity input, the units sequence with the Leray inclusion, the kernel identification, and both component cases are all retained verbatim.

Editorial note. The label lemma-glue-points names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Surjectivity for birational finite morphisms. Varieties, Lemma varieties-lemma-surjective-pic-birational-finite. Gives surjectivity of Pic(X) -> Pic(X') for the glueing morphism.
- The units exact sequence of a glueing. Same chapter, Example example-glue-points; Cohomology, Lemma cohomology-lemma-Leray. Supplies 0 -> O_X^* -> f_*O_{X'}^* -> x_*k^* -> 0 with the ab^{-1} map, and the Leray bound H^1(X, f_*O_{X'}^*) subset H^1(X', O_{X'}^*).
- Regular functions on proper reduced schemes. Varieties, Lemma varieties-lemma-proper-geometrically-reduced-global-sections. Identifies global regular functions on a reduced connected proper scheme with k over an algebraically closed field, forcing ab^{-1} to vanish in the same-component case.

Source reference. Original lemma 01F2: sources/stacks-chow/tags.txt, source lines 1855–1855. Prerequisite pointer invoked at proof line(s) 3350; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0C1J: sources/stacks-curves/pinned-original-source.tex, source lines 3202–3202. Prerequisite pointer invoked at proof line(s) 3325; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0BUG: sources/stacks-chow/tags.txt, source lines 14874–14874. Prerequisite pointer invoked at proof line(s) 3358; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Date: stacks-project:curves.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 0C0T: sources/stacks-chow/tags.txt, source lines 15098–15098. Prerequisite pointer invoked at proof line(s) 3337; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let k be an algebraically closed field. If $f : X' \to X$ is the glueing of two points a, b as in Example 0C1J, then there is an exact sequence*

$$k^* \to \mathrm{Pic}(X) \to \mathrm{Pic}(X') \to 0$$

The first map is zero if a and b are on different connected components of X' and injective if X' is proper and a and b are on the same connected component of X'.

Proof. The map $\mathrm{Pic}(X) \to \mathrm{Pic}(X')$ is surjective by Varieties, Lemma 0C0T. Using the short exact sequence

$$0 \to \mathcal{O}_X^* \to f_*\mathcal{O}_{X'}^* \xrightarrow{ab^{-1}} x_*k^* \to 0$$

we obtain

$$H^0(X', \mathcal{O}_{X'}^*) \xrightarrow{ab^{-1}} k^* \to H^1(X, \mathcal{O}_X^*) \to H^1(X, f_*\mathcal{O}_{X'}^*)$$

We have $H^1(X, f_*\mathcal{O}_{X'}^*) \subset H^1(X', \mathcal{O}_{X'}^*)$ (for example by the Leray spectral sequence, see Cohomology, Lemma 01F2). Hence the kernel of $\mathrm{Pic}(X) \to \mathrm{Pic}(X')$ is the cokernel of $ab^{-1} : H^0(X', \mathcal{O}_{X'}^*) \to k^*$. If a and b are on different connected components of X', then ab^{-1} is surjective. Because k is algebraically closed any regular function on a reduced connected proper scheme over k comes from an element of k, see Varieties, Lemma 0BUG. Thus ab^{-1} is zero if X' is proper and a and b are on the same connected component. □